Dr. Chao YIN

cyinac@connect.ust.hk | pointcloudyc.github.io | GitHub | Google Scholar | ORCID

Research Interests

3D point cloud processing; semantic / instance segmentation and 3D scene understanding; weakly-supervised, long-tailed, and unsupervised learning; **multi-modal AI** (foundation models – SAM, DINO, CLIP – and 2D/3D LLMs); large-scale dataset and benchmark construction; **remote sensing and GIS** applications; scan-to-BIM.

Publications

Preprints (* denotes corresponding author). For the complete and up-to-date list, see Google Scholar.

- **C. Yin**, H. Yue, Q. Han, D. Hu, Z. Liang, F. Lin, B. Sun, B. Wang, M. Li, W. Yao, and J. C. P. Cheng, "Industrial3D: A Terrestrial LiDAR Point Cloud Dataset and Cross-Paradigm Benchmark for Industrial Infrastructure," *arXiv:2603.28660*, 2026.
- **C. Yin**, Q. Han, Z. Hou, Y. Liu, A. Dai, H. Hu, J. Yang, and W. Yao, "Resolving Primitive-Sharing Ambiguity in Long-Tailed Industrial Point Cloud Segmentation via Spatial Context Constraints," *arXiv:2601.19128*, 2026.

Journal Papers

- S. Jing and **C. Yin***, "From Geometric Labels to Semantic Understanding of Indoor Building Components Using Multimodal Large Language Models," *Automation in Construction*, vol. 190, p. 107117, 2026. (JCR Q1, IF: 12.7)
- Q. Han, Z. Wang, **C. Yin***, Z. Hou, and T. Yao, "Semi-Supervised AI for Architectural Heritage Classification and Style Lineage Discovery in Chinese Traditional Settlements," *ISPRS International Journal of Geo-Information*, vol. 15, no. 5, p. 221, 2026.
- B. Wang, Z. Chen, M. Li, Q. Wang, **C. Yin**, and J. C. P. Cheng, "Omni-Scan2BIM: A ready-to-use Scan2BIM approach based on vision foundation models for MEP scenes," *Automation in Construction*, vol. 162, p. 105384, 2024. (JCR Q1, IF: 12.7)
- **C. Yin**, B. Yang, J. C. P. Cheng*, V. J. L. Gan*, B. Wang, and J. Yang, "Label-efficient semantic segmentation of large-scale industrial point clouds using weakly supervised learning," *Automation in Construction*, vol. 148, p. 104757, 2023. (JCR Q1, IF: 12.7)
- D. Hu, V. J. L. Gan, and **C. Yin**, "Robot-assisted mobile scanning for automated 3D reconstruction and point cloud semantic segmentation of building interiors," *Automation in Construction*, vol. 152, p. 104949, 2023. (JCR Q1, IF: 12.7)
- **C. Yin**, J. C. P. Cheng*, B. Wang, and V. J. L. Gan*, "Automated Classification of Piping Components from 3D LiDAR Point Clouds using SE-PseudoGrid," *Automation in Construction*, vol. 139, p. 104300, 2022. (JCR Q1, IF: 12.7)
- Q. Han, **C. Yin***, Y. Deng, and P. Liu, "Towards Classification of Architectural Styles of Chinese Traditional Settlements Using Deep Learning: A Dataset, a New Framework, and Its Interpretability," *Remote Sensing*, vol. 14, no. 20, 2022. (JCR Q2, IF: 5.3)
- B. Wang, Q. Wang, J. C. P. Cheng, C. Song, and **C. Yin**, "Vision-assisted BIM reconstruction from 3D LiDAR point clouds for MEP scenes," *Automation in Construction*, vol. 133, p. 103997, 2022. (JCR Q1, IF: 12.7)
- B. Wang, Q. Wang*, J. C. P. Cheng*, and **C. Yin**, "Object verification based on deep learning point feature comparison for scan-to-BIM," *Automation in Construction*, vol. 142, p. 104515, 2022. (JCR Q1, IF: 12.7)
- **C. Yin**, B. Wang, V. J. L. Gan, M. Wang, and J. C. P. Cheng*, "Automated semantic segmentation of industrial point clouds using ResPointNet++," *Automation in Construction*, vol. 130, p. 103874, 2021. (JCR Q1, IF: 12.7)
- B. Wang, **C. Yin**, H. Luo, J. C. P. Cheng, and Q. Wang, "Fully automated generation of parametric BIM for MEP scenes based on terrestrial laser scanning data," *Automation in Construction*, vol. 125, p. 103615, 2021. (JCR Q1, IF: 12.7)

Conference Papers

- **C. Yin**, B. Wang, and J. C. P. Cheng, "Deep learning-based scan-to-BIM framework for complex MEP scene using laser scanning data," ICCBEI 2019. **Best Paper Award**.

- **C. Yin**, J. C. P. Cheng*, B. Wang, and V. J. L. Gan, "Automated Classification of Piping Components from 3D LiDAR Point Clouds using Deep Learning and Squeeze-Excite Mechanism," IPC 2022.

Book Chapter

- **C. Yin**, B. Wang, K. Chen, H. Liu, C. Huang, J. C. P. Cheng*, and L. Chen, "Digital modelling layer: two digital modelling methods," in *Digital Twins in the Built Environment: Fundamentals, principles and applications*, ICE Publishing, 2022, pp. 101–159.

Education

Hong Kong University of Science and Technology (HKUST) — Hong Kong
Ph.D., Point Cloud Processing, Deep Learning, and BIM Modeling (Advisor: Prof. Jack Cheng) — *Aug 2018 – Aug 2022*

Wuhan University — Hubei, China
M.Phil., Cartography and GIS (Advisor: Prof. Zhongliang Fu) — *Sep 2011 – Jun 2013*

Anhui University of Science and Technology — Anhui, China
B.S., Geographic Information System (GIS) — *Sep 2007 – Jun 2011*

Research & Professional Experience

Geography Institute of Guangzhou, Guangdong Academy of Sciences — Guangzhou, China
Research Scientist (Associate Researcher) — *Sep 2022 – 2026*

- Lead research on weakly-supervised 3D point cloud segmentation, long-tailed learning, multi-modal perception, and remote sensing applications.
- Built **Industrial3D**, the largest terrestrial LiDAR dataset and cross-paradigm benchmark for industrial MEP scene understanding.

Hong Kong University of Science and Technology — Hong Kong
Research Assistant, BIM Lab — *Aug 2018 – Aug 2022*

- Developed deep learning methods for 3D point cloud processing, semantic segmentation, and BIM reconstruction from LiDAR data.
- Key participant in the HK ITF project *"Automated BIM Generation using UAV and Indoor 3D Laser Scanning Technologies"* (PI: Prof. Jack C.P. Cheng); work led to the ICCBEI 2019 **Best Paper Award**.

Hong Kong Polytechnic University — Hong Kong
Visiting Fellow, Lab of Smart City — *Aug 2017 – Jun 2018*

- Key participant in the HK ITF project *"3D Geodatabase Framework for Hong Kong – A Lightweight 3D Seamless Spatial Data Acquisition System (SSDAS)"* (PI: Prof. Wenzhong Shi).

Hengyang Normal University — Hengyang, China
Lecturer, GIS — *Jul 2013 – Jul 2018*

- Taught GIS development courses: C#/.NET programming, ArcGIS Engine, Web GIS, and GIS Software Design and Development.

Research Funding & Projects

Guangdong Province Overseas Postdoctoral Talent Support Program (PI) — Jun 2023 – 2025

- Weakly-supervised semantic segmentation for complex indoor 3D perception using point clouds.

China Postdoctoral Science Foundation (PI) — Sep 2023 – Dec 2024

- Weakly-supervised semantic segmentation for 3D point clouds based on deep long-tail learning.

National Natural Science Foundation of China (Co-PI) — Jan 2021 – Dec 2023

- Automated identification and extraction of landscape genes for Chinese traditional settlements based on 3D semantic models.

Hong Kong ITF (Key Participant; PI: Prof. Jack C.P. Cheng, HKUST) — Oct 2018 – Dec 2021

- **HK$3.4M**. Automated BIM generation using UAV and indoor 3D laser scanning technologies.

Hong Kong ITF (Key Participant; PI: Prof. Wenzhong Shi, PolyU) — Aug 2017 – Jun 2018

- **HK$6.06M**. 3D Geodatabase Framework for Hong Kong – A Lightweight 3D Seamless Spatial Data Acquisition System (SSDAS).

Technical Skills

3D Vision & Point Clouds: PointNet++, KPConv, PTv3, ResPointNet++; semantic/instance segmentation; weakly-supervised, unsupervised, and long-tailed learning

Foundation Models & LLMs: SAM, DINO, CLIP; LLaVA, LISA; LLaMA with LoRA/QLoRA; RAG; 3D multi-modal LLMs

Programming & MLOps: Python (expert, 6+ open-source projects), PyTorch, TensorFlow, Hugging Face, PointCept, Docker, Git

Geospatial & RS: ArcGIS, QGIS, GDAL, CloudCompare, PCL, FME, Revit; Linux, CUDA, HPC clusters; LaTeX

Academic Service

Peer Reviewer: Automation in Construction, ISPRS Journal of Photogrammetry and Remote Sensing, Science of Remote Sensing, Geo-spatial Information Science, Results in Engineering, Remote Sensing

Committee Member: LiDAR Professional Committee, China National Committee for the International Society for Digital Earth (CNISDE)

Open-Source Contributions: Active maintainer of 6+ research code repositories with **140+ GitHub stars** (github.com/PointCloudYC); all published papers ship publicly available code for reproducibility.

Awards & Honors

Best Paper Award, International Conference on Construction, Building Engineering and Innovation (ICCBEI), 2019.

Last updated: Jul 6, 2026